

MADOI, China

WMO: 560330

Lat: 34.9172N

Lon: 98.2158E

Elev: 4273

StdP: 59.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-24.1	-34.7	0.2	-18.1	-32.7	0.3	-16.9	13.1	-6.1	11.7	-5.6	0.7	40	0.857

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	17.5	7.8	15.7	7.0	14.3	6.4	9.3	15.0	8.6	13.8	7.7	12.5	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	10.7	11.7	6.3	10.2	11.0	5.6	9.7	10.4	40.8	15.0	38.4	14.0	36.0	12.5	13.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.1	7.8	DB	-30.1	20.5	2.1	1.6	-31.6	21.7	-32.8	22.6	-33.9	23.4	-35.5	24.6	(4)
(5)			WB	-30.4	10.6	2.0	0.9	-31.8	11.3	-33.0	11.8	-34.1	12.2	-35.6	12.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	-14.4	-11.2	-7.1	-1.6	2.6	6.1	8.7	8.6	5.1	-1.5	-8.8	-13.3	(6)	
	DBStd	8.79	3.69	3.64	3.61	3.10	2.64	2.56	2.55	2.74	3.05	3.46	3.51	3.57	(7)	
	HDD10.0	4485	755	592	531	349	231	119	56	60	149	357	563	723	(8)	
	HDD18.3	7487	1013	826	789	599	489	367	297	300	397	615	813	982	(9)	
	CDD10.0	40	0	0	0	0	0	3	17	18	2	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	3.0	2.5	3.0	3.4	3.5	3.5	3.4	3.3	3.2	2.9	2.8	2.5	2.3	(14)	
	PrecAvg	309	3	4	7	10	29	57	71	61	47	17	3	2	(15)	
(16) Precipitation	PrecMax	492	11	11	21	24	61	120	147	152	123	45	9	9	(16)	
	PrecMin	184	0	0	0	2	2	7	35	13	11	3	0	0	(17)	
	PrecStd	69	3	3	5	5	12	25	26	33	24	10	3	2	(18)	
	PrecStd	69	3	3	5	5	12	25	26	33	24	10	3	2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	3.2	7.2	11.5	14.6	16.8	19.9	20.1	16.9	12.4	3.5	0.6	(19)	
		MCWB	-6.9	-4.9	-2.7	0.2	3.2	7.1	9.4	8.2	7.4	3.2	-4.2	-6.2	(20)	
	2%	DB	-2.8	0.2	4.1	9.0	12.0	14.6	17.5	17.6	14.3	8.4	1.6	-2.1	(21)	
		MCWB	-8.3	-6.6	-4.0	-0.9	2.0	5.9	8.1	7.8	6.3	1.0	-5.1	-8.0	(22)	
	5%	DB	-5.0	-1.7	2.0	7.0	10.1	13.0	15.8	15.7	12.4	5.9	-0.1	-4.1	(23)	
		MCWB	-9.6	-7.6	-5.1	-1.8	1.1	4.9	7.2	7.3	5.6	-0.4	-6.0	-9.2	(24)	
	10%	DB	-7.1	-3.6	0.1	5.0	8.3	11.4	14.1	14.0	10.4	4.1	-1.8	-6.0	(25)	
		MCWB	-11.0	-8.6	-5.9	-2.7	0.5	4.4	6.9	6.6	4.5	-1.3	-6.8	-10.3	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.4	-4.7	-2.0	1.4	5.1	9.1	10.7	10.2	8.8	4.7	-3.4	-5.9	(27)	
		MCDB	-0.8	2.1	5.2	8.5	11.0	14.5	17.2	16.3	14.1	10.4	2.2	0.0	(28)	
	2%	WB	-8.2	-6.3	-3.6	0.0	3.6	7.8	9.6	9.1	7.5	2.2	-4.5	-7.8	(29)	
		MCDB	-3.3	-0.4	2.7	6.1	8.9	12.7	15.3	14.9	12.6	6.6	0.4	-2.5	(30)	
	5%	WB	-9.5	-7.3	-4.6	-1.1	2.5	6.7	8.7	8.4	6.5	0.8	-5.5	-9.0	(31)	
		MCDB	-5.4	-2.2	1.1	5.2	7.7	11.0	13.9	13.8	11.1	4.4	-0.9	-4.5	(32)	
	10%	WB	-10.8	-8.4	-5.7	-2.0	1.6	5.6	7.9	7.6	5.5	-0.6	-6.5	-10.2	(33)	
		MCDB	-7.3	-4.0	-0.5	4.0	6.4	9.5	12.4	12.5	9.5	3.1	-2.2	-6.2	(34)	
(36) Mean Daily Temperature Range	MDBR		13.9	13.6	12.9	11.7	10.2	9.1	9.4	9.7	9.5	9.9	12.8	14.1	(35)	
	5% DB	MCDBR	16.0	16.0	15.4	14.6	13.6	11.7	12.0	11.9	11.5	12.0	14.2	16.0	(36)	
		MCWBR	11.7	10.8	9.3	7.2	6.1	5.1	4.9	4.8	5.2	6.1	8.9	11.4	(37)	
	5% WB	MCDBR	15.3	15.1	13.9	12.6	10.3	9.4	9.5	9.4	9.6	10.1	13.0	15.5	(38)	
		MCWBR	11.3	10.4	8.5	6.7	5.3	4.8	4.4	4.2	4.7	5.7	8.5	11.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.186	0.214	0.264	0.300	0.310	0.295	0.284	0.278	0.259	0.236	0.198	0.180	(40)	
	taud		2.571	2.490	2.340	2.257	2.320	2.467	2.561	2.571	2.599	2.589	2.587	2.613	(41)	
	Ebn at Noon		1072	1058	1015	983	972	981	990	996	1010	1018	1046	1066	(42)	
	Edh at Noon		74	91	117	135	129	111	100	97	90	82	72	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.27	4.06	5.18	6.18	6.40	6.28	6.48	6.03	5.05	4.32	3.72	3.14	(44)	
	RadStd		0.28	0.27	0.29	0.26	0.38	0.43	0.41	0.46	0.40	0.23	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.41	N/A	N/A	N/A	N/A	N/A	-133	-160	N/A	N/A
(47) Regional (2 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	-110	-128	N/A	N/A

Nomenclature: See separate page